

Khulna University of Engineering & Technology  
B.Sc. Engineering 3<sup>rd</sup> Year 1<sup>st</sup> Term Examination, 2020  
Department of Materials Science and Engineering

**MSE 3101**  
**(Transport Phenomena in Materials)**

Time: 1.5 hours

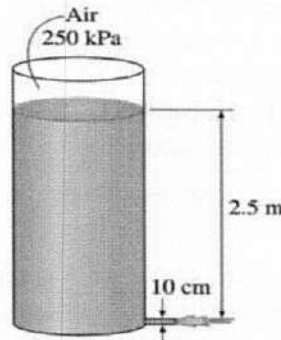
Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks  
(iii) Assume reasonable data if any missing

**Section A**

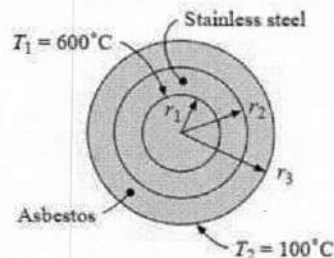
(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Write down the three major assumptions used in the derivation of the Bernoulli equation and derive the Bernoulli equation. (20)
- (b) A pressurized tank of water has a 10 cm diameter orifice at the bottom, where water discharges to the atmosphere. The water level is 2.5 m above the outlet [Fig: 1 (b)]. The tank air pressure above the water level is 250 kPa (absolute) while the atmospheric pressure is 100 kPa. Neglecting frictional effects, determine the inertial discharge rate of water from the tank. (10)



**Figure: 1(b)**

- Q 2. (a) Consider a slab of thickness  $L$  and constant thermal conductivity  $k$  in which energy is generated at a constant rate of  $g_0$   $W/m^3$ . The boundary surface at  $x = 0$  is insulated (adiabatic) and that at  $x = L$  dissipates heat by convection with a heat transfer coefficient  $h$  into a fluid at a temperature  $T_\infty$ . Develop expressions for the temperature  $T(x)$  and the heat flux  $q(x)$  in the slab. (15)
- (b) A thick-walled tube of stainless steel [18 %Cr, 8 %Ni,  $k = 19$   $W/(m \cdot ^\circ C)$ ] with 2 cm inner diameter and 4 cm outer diameter is covered with a 3 cm layer of asbestos insulation  $k = 0.2$  ( $W/m \cdot ^\circ C$ ). If the inside wall temperature of the pipe is maintained at  $600^\circ C$ , calculate the heat loss per meter of length. [Fig: 2 (b)] (15)



**Figure: 2(b)**

- Q 3. (a) An aluminium plate [ $k = 150$   $W/(m \cdot ^\circ C)$ ,  $\rho = 2790$   $Kg/m^3$ ,  $C_p = 0.88$   $kJ/(Kg \cdot ^\circ C)$ ] of thickness  $L = 3$  cm and at a uniform temperature of  $T_o = 225^\circ C$  is suddenly immersed at time  $t = 0$  in a well stirred fluid maintained at a constant temperature  $T_\infty = 23^\circ C$ . The heat (15)

- transfer coefficient between the plate and the fluid is  $h = 320 \text{ W}/(\text{m}^2 \cdot ^\circ\text{C})$ . Determine the time required for the center of the plate to reach  $45^\circ\text{C}$ .
- (b) Calculate the critical radius of insulation for asbestos [ $k = 0.17 \text{ W}/(\text{m} \cdot ^\circ\text{C})$ ] surrounding a pipe and exposed to room air at  $20^\circ\text{C}$  with  $h = 3.0 \text{ W}/(\text{m}^2 \cdot ^\circ\text{C})$ . Calculate the heat loss from a  $200^\circ\text{C}$ ,  $5.0 \text{ cm}$  diameter pipe when covered with the critical radius of insulation and without insulation. (15)

### Section B

(Answer ANY TWO questions from this section in answer script B)

- Q 4. (a) Species A is evaporating from a flat surface into species B. Assume that the concentration profile for species A in the concentration boundary layer is of the form  $C_A(y) = Dy^2 + Ey + F$ , where  $D, E$  and  $F$  are constants at any  $x$ -location and  $y$ - is measured along a normal from the surface. Develop an expression for the mass transfer convection coefficient  $h_m$  in terms of these constants, the concentration of A in the free stream  $C_{A,\infty}$  and the mass diffusivity  $D_{AB}$ . (15)
- (b) Air at a free stream temperature of  $T_\infty = 20^\circ\text{C}$  is in parallel flow over a flat plate of length  $L = 5\text{m}$  and temperature of  $T_s = 90^\circ\text{C}$ . However, obstacles placed in the flow intensify mixing with increasing distance  $x$  from the leading edge and the spatial variation of temperatures measured in the boundary layer is correlated by an expression of the form  $T(^\circ\text{C}) = 20 + 70 \exp(-600xy)$ , where  $x$  and  $y$  are in meters. Determine and plot the manner in which local convection coefficient  $h$  varies with  $x$ . Evaluate the average convection coefficient  $\bar{h}$  for the plate. Given that for  $328 \text{ K}$ ,  $k = 0.0284 \text{ W}/(\text{m} \cdot \text{K})$ . (15)
- Q 5. (a) Derive an expression for the log mean temperature difference (LMTD) for a counter flow heat exchanger. (13)
- (b) A counter flow concentric tube heat exchanger used for engine cooling has been in service for an extended period of time. The heat transfer surface area of the exchanger is  $5 \text{ m}^2$  and the design value of the overall convection coefficient is  $38 \text{ W}/(\text{m}^2 \cdot \text{K})$ . During a test run, engine oil flowing at  $0.1 \text{ Kg/s}$  is cooled from  $110^\circ\text{C}$  to  $66^\circ\text{C}$  by water supplied at a temperature of  $25^\circ\text{C}$  and a flow rate of  $0.2 \text{ Kg/s}$ . Determine whether fouling has occurred during the service period. If so, calculate the fouling factor,  $R''_f(\text{m}^2 \cdot \text{K}/\text{W})$ . Given that for engine oil ( $\bar{T}_h = 361 \text{ K}$ ),  $C = 2166 \text{ J}/\text{Kg} \cdot \text{K}$  and for water ( $\bar{T}_c = 304 \text{ K}$ ),  $C = 4178 \text{ J}/\text{Kg} \cdot \text{K}$ . (17)
- Q 6. (a) Derive an expression for radiation intensity and its relation to emission. (20)
- (b) A small surface of area  $A_1 = 10^{-3} \text{ m}^2$  is known to emit diffusely and from measurements the total intensity associated with emission in the normal direction is  $I_n = 7000 \text{ W}/\text{m}^2 \cdot \text{sr}$ . Radiation emitted from the surface is intercepted by three surfaces of area  $A_2 = A_3 = A_4 = 10^{-3} \text{ m}^2$ , which are  $0.5 \text{ m}$  from  $A_1$  and are oriented as shown in Fig: 6 (b). (10)

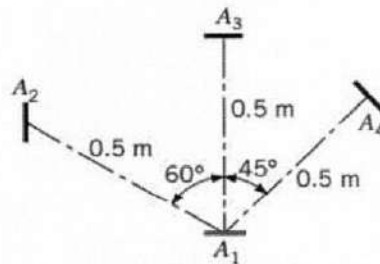


Figure: 6(b)

What is the intensity associated with emission in each of the three directions? What are the solid angles subtended by the three surfaces when viewed from  $A_1$ . What is the rate of which radiation emitted by  $A_1$  is intercepted by the three surfaces?

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**MSE 3103**  
**(Materials Manufacturing Processes)**

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks  
(iii) Assume reasonable data if any missing

**Section A**

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) What do you mean by the term "undercooling"? Examine how the depression of (12)  
undercooling influence the solidified structure of pure metals and alloys.
- (b) Write short notes on the mechanisms of clay-water bonding in green molding sand (12)  
aggregates.
- (c) For a homogeneous nucleation, undercooling of the order of 270°C is required for liquid (06)  
aluminum. Determine the size of the critical radius of the spherical embryo and the energy  
change associated with the process. If the volume of one aluminum atom is about  $1.25 \times 10^{-23} \text{ cm}^3$ ,  
calculate the probable number of copper atoms required to form such embryo.  
Given data:  $T_e = 660^\circ\text{C}$ ,  $\gamma_{sl} = 135 \text{ erg/cm}^2$ ,  $L = 1.47 \times 10^{10} \text{ erg/cm}^3$ .
- Q 2. (a) Analyze why dendrites are formed during solidification of metal alloys but not of pure (12)  
metals. Examine the rule of DAS in controlling the mechanical properties of cast materials.
- (b) Deduce an expression for the determination of core print length X of a core having diameter (12)  
D and length L. A hollow steel component of having 20mm inner diameter and 40mm  
length to be cast. Determine the length of the core print. Given: Steel density  $7.8 \text{ g/cm}^3$ ,  
core density  $1.67 \text{ g/cm}^3$ , core compressive strength  $2.5 \text{ g/cm}^2$ .
- (c) How do you create quality casting? (06)
- Q 3. (a) A thin-wall cylindrical casting of 100mm inside diameter, 10mm thickness and 250mm  
length is to be cast using cast iron in greensand mould. Determine the followings-
- i) Using 20% longer than solidification time, calculate the pouring time required for the (05)  
casting. The mould constant in Chvorinov's rule is  $1.2 \text{ s/mm}^2$ .
- ii) Using a 50mm high-pouring basin, calculate the effective metal head pressure developed (05)  
in the gating system. List all assumptions you have made during this calculation.
- (iii) Calculate the choke area of the gating system. Use  $7.0 \text{ g/cc}$  for liquid metal density and (05)  
 $0.80$  for discharge coefficient.
- (iv) Calculate sprue top and bottom areas, runner area and gate area of the gating system. (15)  
Use critical velocity =  $500 \text{ mm/s}$  for this unpressurised gating system with a gating ratio =  
 $1 : 2 : 4$ . Also use a safety factor of 25% to oversize the sprue top area.

### **Section B**

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) What is an orthogonal cutting operation? Name and briefly describe the four types of chips that occur in metal cutting. (12)
- (b) What are the advantages of machining? What properties should cutting tools have? (08)
- (c) What do you understand by sheet metal forming? What do you produce by using this process? (10)
- Q 5. (a) Describe in words what the Merchant equation tells us. (08)
- (b) 'Tempered glass is much more resistant to scratching and breaking'-explain the reason of such strength. (10)
- (c) What are the general properties of ceramics? Classify the body materials of ceramic and their function. (12)
- Q 6. (a) Why additive manufacturing (AM) is so popular now? Write down the advantages of different AM techniques. (10)
- (b) Which one is technically more feasible and economical for double-layer bottles making of polymer between extrusion blow molding and injection blow molding? Justify your answer. (12)
- (c) Draw a process flow chart of powder metallurgy. (08)

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**MSE 3105**  
**(Welding and Materials Joining Processes)**

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks

**Section A**

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) Define metal joining process. With appropriate diagrams, explain the adhesive-bonded joint configurations. (12)  
(b) How does molten metal bath dip brazing differ from molten chemical (flux) bath dip brazing? (12)  
(c) What properties do soldering fluxes must possess? (06)
- Q 2. (a) Explain the process of electroslog welding with neat sketches. (10)  
(b) Compare and contrast the shielded-metal arc welding with the gas-metal arc welding. (15)  
(c) Write down the names of weld joints with appropriate diagrams. (05)
- Q 3. (a) Explain the working principle of laser beam welding with neat sketches. (13)  
(b) How does friction welding differ from diffusion welding? (10)  
(c) How does the resistance welding process work? (07)

**Section B**

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Discuss the weld metal nucleation mechanisms. (10)  
(b) Explain the variation of growth mode from fusion boundary to the weld centerline due to the variation of temperature gradient, G and growth rate, R along weld pool boundary. (10)  
(c) At the welding speed of 1000 mm/min straight columnar grains point toward the centerline, while at 250 mm/min curved columnar grains point in the welding direction. Explain the formation of these structures with figure. (10)
- Q 5. (a) Discuss the problems associated with the partially melted zone in welding. How can these problems be overcome? (15)



(b) Fig. 5(b) shows the hardness profiles of a 3.2-mm-thick 6061 aluminium welded in T6 (10+5) condition (right after welding, postweld natural aging for 7 days and postweld artificial aging at 155°C for 18 hours).

- (i) Explain the existence of hardness minimum after both aging processes and recovery of full hardness near fusion boundary due to artificial aging.
- (ii) Is there any method to recover full strength in the HAZ? Explain.

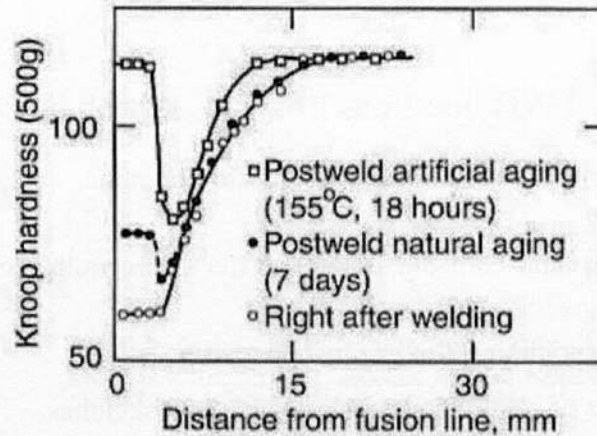


Figure 5(b)

- Q 6. (a) "Welding of higher carbon steels is more difficult than welding of lower carbon steels." (08)  
- Why?
- (b) What is lamellar tearing in a weldment? How does it form? How can lamellar tearing be avoided by modifying the design of weldments? (10)
- (c) How can the internal defects in a weldment be identified by NDT? Discuss the principles of any NDT method to identify internal defects in weldments. (12)

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**MSE 3107**  
**(Physical Metallurgy of Materials)**

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.  
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(iii) Assume reasonable data if any missing

**Section A**

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) What is continuous cooling transformation (CCT) diagram? Draw and explain the CCT diagram for a typical eutectoid steel along with slow to fast cooling and mentioned different microstructure of steel. (14)
- (b) Explain normalizing and annealing process. Why normalizing is better than annealing process? (08)
- (c) Calculate normalizing temperature of the following hypereutectoid steel. (08)

| AISI-SAE Number | % C | % Mn | % Si | % Cr |
|-----------------|-----|------|------|------|
| 52100           | 1.0 | 0.30 | 0.25 | 1.30 |

- Q 2. (a) What are the effects of phosphorus and silicon on the properties of steel? (07)
- (b) What is martensite? How BCT structure is formed from BCC structure? Explain with necessary figures. (13)
- (c) Determine the carbon-equivalent (CE) of the following steels and comments on heating rate and sensitivity. (10)

| AISI-SAE Number | % C  | % Mn | % Si | % Ni | % Cr | % Mo | % Y  |
|-----------------|------|------|------|------|------|------|------|
| 4340            | 0.40 | 0.60 | 0.30 | 2.0  | 0.70 | 0.20 | --   |
| 8620            | 0.20 | 0.70 | 0.15 | 0.40 | 0.60 | --   | 0.20 |

- Q 3. (a) Explain the effect of carbon contain on the properties of steel with figures. (08)
- (b) Write different types of annealing? Explain with necessary figures. (14)
- (c) Calculate the full annealing temperature for the steels of the following grades (08)

| AISI-SAE Number | % C  | % Mn | % Si | % Cr | % Mo |
|-----------------|------|------|------|------|------|
| 1095            | 0.90 | 0.30 | --   | --   | --   |
| 4140            | 0.40 | 0.80 | 0.30 | 0.80 | 0.20 |

### **Section B**

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) Compare between pack and gas carburizing. Which of them is more economical for mass production? (10)
- (b) What do you mean by surface alloying? Briefly discuss the mechanism of the process with necessary figures. (10)
- (c) How do the mechanical properties of steel vary with different tempering temperature? (10)
- Q 5. (a) Explain the precipitation strengthening mechanism of Al-Cu alloy with a schematic phase diagram. (15)
- (b) What is thermo-mechanical treated alloy? Discuss the microstructure of this alloy with proper example. (10)
- (c) What do you mean by ultra-high strength steels and micro-alloyed steels? (05)
- Q 6. (a) What is the necessity of heat treatment of Al alloys? Discuss the solution heat treatment process of Al alloy with proper example. (15)
- (b) What are the criteria required for an alloy to be called superalloy? Briefly discuss the major phases present Ni based super alloy. (15)



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**Hum 3127**  
**(Government and Sociology)**

Time: 1.5 hours

Full Marks: 120

- N.B. (i) Answer **ANY TWO** questions from each section in separate scripts.  
(ii) Figure in the right margin indicates full marks

**Section A**

(Answer **ANY TWO** questions from this section in **answer script A**)

- Q 1. (a) What do you mean by politics? People do say that politics is a dirty game, is it true? (15)  
Discuss.
- (b) Define state and government. What are the differences between state and government? (15)
- Q 2. (a) What is meant by democracy? Critically explain the essential conditions for the successful working of the democratic country. (15)
- (b) What are the main features of dictatorship? What are its merits and demerits? (15)
- Q 3. (a) What is constitution? How can we say constitution is fundamental and organic law of the nation? (15)
- (b) What are the methods of securing the independence of Judiciary? (15)

**Section B**

(Answer **ANY TWO** questions from this section in **answer script B**)

- Q 4. (a) What do you mean by social stratification? (10)
- (b) Discuss major types of social stratification. (20)
- Q 5. (a) What is social structure? (10)
- (b) Describe the Marx's concept of social structure. (20)
- Q 6. (a) What do you mean by culture? (05)
- (b) Mention the characteristics of culture. (10)
- (c) Describe the basic concept of cultural lag. (15)